

Ryan DUNG

+33 7 83 39 68 91 | dungryan@outlook.fr | Champigny-sur-Marne, Île-de-France, France
linkedin.com/in/ryandung | github.com/Ryqn0 | kaggle.com/ryqn00 | ryqn0.github.io

PROFILE

Engineering graduate from IMT Nord Europe, specialized in data science, MLOps and optimization. Hands-on experience taking projects end-to-end, from data preparation to ML model deployment in production on the cloud (GCP). Seeking a first position as a **Data Scientist, AI/ML Engineer** or **Data Engineer**.

PROFESSIONAL EXPERIENCE

Data Analyst / Data Scientist – Procurement

Sep 2025 – Mar 2026

COLAS (BOUYGUES Group)

Paris, France

- Designed and maintained Power BI dashboards powered by SQL Server (SQL queries, data modeling)
- Cleaned and prepared data, then trained classifiers in Python (classical ML, neural networks, LLMs)
- Applied operations research and optimization to tender allocation problems

PROJECTS

RAG Agent for the French Financial Markets Authority (AMF)

github.com/Ryqn0/amf-rag-agent

RAG-based AI chatbot

- Conversational agent answering questions on the financial data of French companies, grounded in documents published by the AMF
- Ingestion pipeline for PDF and HTML documents (French/English): chunking, embedding with sentence-transformers, then vector storage in a ChromaDB database
- Bilingual interface built with Streamlit; agent powered by the Anthropic API (Claude Haiku), workflow built with LangGraph and observability via LangStudio, containerized with Docker and deployed on Hugging Face Spaces and monitoring with Prometheus + Grafana
- Demo: huggingface.co/spaces/Ryqn/amf-rag-agent

Tools: Python, LangGraph, LangStudio, Anthropic API, FastAPI, Streamlit, ChromaDB, sentence-transformers, Docker, Pytest, Git, Hugging Face Spaces, Prometheus + Grafana

Real-Time Banking Fraud Detection

github.com/Ryqn0/fraud-mlops

Apr – May 2026

End-to-end MLOps pipeline on GCP

- Real-time streaming: Pub/Sub → Cloud Run (ingestion + FastAPI inference) → BigQuery, with sliding-window SQL feature engineering and LightGBM training using a leakage-safe temporal split
- Data drift monitoring (Kolmogorov-Smirnov test for continuous variables, χ^2 for binary) with Cloud Monitoring alerts; MLflow experiment tracking; AUC-PR measured on a temporal split
- CI/CD with GitHub Actions: 21 unit and behavioral tests, automatic build and deployment of both Cloud Run services on merge to main
- Cost-controlled (under \$50) via automation scripts (provisioning and teardown); 7 orchestrated GCP services (Pub/Sub, Cloud Run, BigQuery, GCS, Artifact Registry, Cloud Monitoring, IAM)

Tools: Python, LightGBM, FastAPI, Pub/Sub, Cloud Run, BigQuery, MLflow, Docker, GitHub Actions, GCS, Cloud Monitoring, Pytest

Customer Churn Prediction API

github.com/Ryqn0/Customer-Churn-S6E3-Challenge-Kaggle-MLOPS

Mar – Apr 2026

Kaggle competition, MLOps

- Containerized prediction API for a Kaggle competition, deployable on Cloud Run (GCP)
- Full pipeline: training, inference, tests, API and CI/CD; ensemble learning via Voting Classifier (XGBoost, LightGBM, CatBoost)

Tools: Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, Great Expectations, MLflow, FastAPI, Gradio, Pytest, Docker, GitHub Actions, Artifact Registry, Cloud Run

Supplier Selection Optimization

Operations Research

Dec 2025 – Jan 2026

Python, PuLP, OR-Tools

- Generalizable model for tender allocation with rebate systems
- Python notebooks processing Excel files to identify optimal scenarios
- 0.5–1 % gain on €10M tenders through heuristics and operations research

Tools: Python, PuLP, OR-Tools, Gurobi, Pandas, OpenPyXL

Multi-Label Classification of Customer Service Comments

LLM Fine-Tuning

Nov 2025

Python, Transformers, Qwen

- Fine-tuned an open-source LLM to summarize and classify customer service comments
- Used generative AI for data augmentation and label annotation
- Fine-tuned Qwen2.5-0.5B with Q-LoRA (Quantized Low-Rank Adaptation), 4-bit quantization

Tools: Python, Hugging Face Transformers, PEFT, bitsandbytes, PyTorch, Qwen2.5

EDUCATION

IMT Nord Europe

Master of Engineering in Computer Science

Douai, France

Sep 2022 – Mar 2026

- Coursework: Numerical Analysis, Probability, Statistics, Optimization, ML, Databases

Shanghai Jiao Tong University

Exchange program, Data Science specialization

Shanghai, China

Sep 2024 – Jun 2025

- Coursework: Data Science, ML, Deep Learning, Data Structures, Mandarin Chinese (HSK5)

Lycée Buffon

Classes Préparatoires PSI (intensive undergraduate program, Physics/Engineering track)

Paris, France

Sep 2021 – Jun 2022

Lycée Fénelon

Classes Préparatoires PCSI (intensive undergraduate program, Physics/Chemistry track)

Paris, France

Sep 2020 – Jun 2021

SKILLS

Languages : Python, SQL, C++, Java, C

ML / Data Science : Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna

Deep Learning / LLM : PyTorch, TensorFlow, Hugging Face Transformers, PEFT (LoRA / Q-LoRA), bitsandbytes

Agentic AI / RAG : LangChain, LangGraph, LangStudio, Anthropic API, ChromaDB, sentence-transformers, Elasticsearch

MLOps : MLflow, Docker, Kubernetes, FastAPI, Pytest, GitHub Actions (CI/CD), Great Expectations, drift monitoring

Cloud – GCP : Cloud Run, Pub/Sub, BigQuery, GCS, Artifact Registry, Cloud Monitoring, IAM

Optimization / OR : PuLP, OR-Tools, Gurobi

Data & BI : Power BI, SQL Server, Databricks, Excel

Spoken languages : French (native), English (fluent), Mandarin Chinese (B2, HSK5)